Fiscal Year:	FY 2004	**Task Last Updated:**	FY 04/03/2006
PI Name:	Schaffler, Mitchell B. Ph.D.		
Project Title:	Resorption suppression and bone health disuse		
Division Name:	Human Research		
Program/Discipline:	NSBRI Teams		
Program/Discipline--Element/Subdiscipline:	NSBRI Teams--Bone Loss Team		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	10029-6500	**Congressional District:**	14
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	NSBRI
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No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
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Performance Goal No.:			
Performance Goal Text:			
	Bone loss in microgravity, and the resulting bone fragility that ensues, have been identified by NASA and NSBRI as key barriers to successful long-term spaceflight and the recovery of normal function in astronauts upon returning to earth’s gravity. Overcoming these problems will require safe and effective countermeasures not only to prevent bone loss, but also to maintain the functional-mechanical integrity of the tissue during prolonged spaceflight. A necessary prerequisite for development of those countermeasures is the identification of appropriate cellular targets for both processes. Those cellular targets have not yet been fully characterized. We posit that 1) preventing bone loss and 2) maintaining bone health during long-duration in the absence of normal loading involve different cellular mechanisms and so will require different countermeasures. 1) Osteoclasts are clearly the agents driving bone loss due to unloading. Thus, they present an obvious countermeasure target for modulating bone loss in spaceflight and in other hypodynamic loading situations. Available pharmacological strategies to inhibit osteoclastic resorption have a high likelihood of		

Task Description:	success, though definitive long-term data for remodeling suppression after unloading do not exist. 2) Changes in osteocyte integrity result from long-term loss of normal mechanical loading. If the normal remodeling response of bone to impaired osteocytes is suppressed, as would be the case with treatment using an anti-resorptive agent, then regions of osteocytes can die, leading to the accumulation of devitalized bone. Devitalized bone becomes mechanically fragile, raising the fundamental question of whether bone loss resulting from withdrawal of normal mechanical usage can be safely prevented for the long-term, without paradoxically impairing osteocyte function, and thereby bone's ability to function mechanically in a normal load-bearing environment. The proposed experiments test the hypotheses that 1) Long-term suppression of bone remodeling in disuse will successfully maintain bone mass, microarchitecture, stiffness and strength, but will result in compromised fracture resistance properties; and 2) Decreased mechanical usage in the presence of an antiresorptive agent results in loss of osteocyte integrity and accumulation of bone with impaired viability. To test these hypotheses, we will undertake a series of long-term immobilization experiments in a canine model, with bisphosphonate treatment to prevent bone loss. Bone health will be assessed from conservation of tissue mechanical properties and from in situ assessments of osteocyte viability. We will determine whether suppression of bone resorption superimposed on unloading leads to impaired osteocyte viability and increased brittleness of bone, and whether the extent of such alterations can be sufficient to cause significant bone fragility.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The long term goal of our program is to define cellular targets for 1) preventing bone loss and 2) maintaining bone health during long-duration spaceflight. We posit that these two processes involve different cellular mechanisms and so will require different countermeasures. That osteoporosis in higher mammals due to loss of normal mechanical loading results principally from osteoclastic bone resorption is well established. Pharmaceutical agents and biophysical strategies to suppress bone resorption have proved effective in preventing disuse bone loss in the short term. Thus, targeting osteoclasts to prevent bone loss due to reduced mechanical loading is an obvious countermeasure strategy and needs to be explored more fully; long-term data showing sustained efficacy are needed. Moreover, we believe that the skeletal problems resulting from long-term disuse are more complex than can be solved by preventing bone resorption alone.
Task Progress:	No progress report this period.
Bibliography Type:	Description: (Last Updated: 08/21/2020)